

A Fact Worth Explaining

Research Design Module 4

Beaver, McNichols, and Wang (2020), *Journal of Accounting and Economics*

ACCTG 597A · Fall 2026 · Sam Bonsall · Penn State

Two numbers, fifteen years apart

Mean one-day market response to an earnings announcement (Beaver's 1968 measure):

2001

3.42

2016

11.70

While parts of the literature worried that financial statements were losing relevance, earnings announcements were becoming three to four times bigger market events.

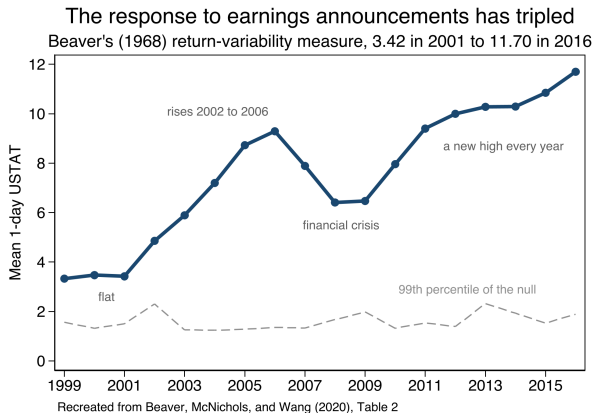
The instrument: Beaver's (1968) measures, modernized

The measure. *USTAT*: squared announcement-day market-model residual, scaled by the average squared residual on non-announcement days. A value of 1 means an announcement day looks like any other day. *AVOL* is the volume analogue.

The upgrade. Time-stamping announcements to trading hours: assigning after-hours announcements to the correct trading day raises measured day-0 information content from 4.08 to 7.59, an 86% difference.

Correct time-stamping nearly doubles the measured response; mis-stamped announcements smear it across days 0 and +1.

The headline fact: a threefold increase since 2001



The fact overturned the question the literature was asking

- Landsman and Maydew (2002) asked whether information content had *declined* over three decades; the concern of the era was preemption and fading relevance
- Francis, Schipper, and Vincent (2002) documented some increase in announcement usefulness through the 1990s, at much smaller magnitudes; Ball and Shivakumar (2008) framed the announcement window as a modest slice of the year's information
- BMW: the post-2001 increase is “an order of magnitude greater” than the increases the 1990s literature documented for three-day price reactions

The interesting question flipped from “why so little information?” to “why so much, and why now?”

Candidate explanation: the announcement itself changed

Earnings announcements increasingly bundle other disclosures (1999 to 2016):

Bundled content	1999	2016
Management guidance	3%	36%
Same-day analyst forecasts	34%	70%
Financial-statement line items disclosed	22%	57%
of which: cash-flow statement items	0.5%	32%

Announcements with each type of bundled content show a higher market response in every sample year.

The bundles explain part of the trend, not all of it

Measure of market response	Share of trend explained
<i>USTAT</i> (return variability)	42%
<i>USTAT</i> numerator alone	nearly all
<i>AVOL</i> (volume)	62%
<i>AVOL</i> numerator alone	≈ 75%

“The concurrent disclosures and forecasts we examine explain from 42% to all of the time trend in increased investor response, depending on the measure of market response. In addition, financial statement line items alone contribute half of this explanatory power.”

BMW (2020), Introduction

It is not that investors respond more to everything

The same disclosure types, issued on days *without* an earnings announcement, move differently (mean *USTAT*, 1999 to 2016):

Stand-alone event	1999	2016	Trend
Earnings announcement	2.77	7.61	rising
Management guidance	11.15	4.50	falling
Analyst forecast	1.43	1.90	roughly flat

The trend differences are statistically significant ($F = 41$ to 48). The increase is specific to information released with earnings, which is consistent with a disclosure-content interpretation rather than a general change in investor responsiveness.

What remains open, and BMW say so

- Roughly 58% of the headline *USTAT* trend is unexplained by the paper's own variables
- *USTAT* and the Ball-Shivakumar R^2 diverge in 2013–2016: “We have no explanation for why the adjusted R^2 declines . . . when the *USTAT* increases” (fn. 14)
- The secular decline in background volatility (the denominator), the regulatory channel, and the post-2016 path are all left to future work

“First and foremost, while our concurrent information variables explain a significant part of the increase in market response over time, our quest for an explanation is not complete.”

BMW (2020), Conclusion

Why this is a model of descriptive research

- A striking, robust fact, documented with serious attention to measurement (time stamps, one-day windows, numerator and denominator examined separately)
- Candidate explanations tested jointly and against a falsification design, with the unexplained residual attributed honestly
- An afterlife consistent with the point: media coverage (Guest 2021), algorithmic trading (Lee and Watts 2020), and a critique of the *USTAT* measure itself (Thomas, Zhang, and Zhu 2022) all followed

The fact has continued to attract both explanations and measurement scrutiny.

Summary

- The one-day market response to earnings announcements roughly tripled between 2001 and 2016; bundled guidance, analyst forecasts, and line-item disclosures explain between 42 percent and nearly all of the trend, depending on the measure.
- The stand-alone-disclosure comparison suggests the increase is specific to the earnings announcement rather than a general change in investor responsiveness to information.
- Precise measurement, explanations tested jointly, and an honestly attributed residual allow descriptive evidence to set a research agenda in the absence of a causal design.

Beaver (1968) · Beaver, McNichols, and Wang (2018, 2020) ·
github.com/sbb151/acctg597a-research-design